

DES: Design

MIS: Mission

PERF: Performance

Design Requirements

V1.00

Note: Rough draft of general requirements needed to operate the ultrasonic sensor, servo motor, and the gyroscope

Tier	REQ-ID	Parent ID	Requirement Text	Verification
1	MIS-01	—	The system shall empty food into a bin when an object is detected in its range of ___ (TBD)	Demonstration
1	MIS-02	—	The program shall open the wall for a second every 30 second, a minute, or 2 minutes based on the potentiometer	Demonstration
1	MIS-03	—	The system shall open the wall when the user points his hands down and answer to specific prompts	Demonstration
1	DES-01	MIS-01	The ultrasonic sensor shall be placed on the top of the design and pointed downwards	Demonstration
1	DES-02	—	The servo motor shall pull on a string that is attached to a cardboard wall	Demonstration
1	PERF-01	MIS-03	A glove shall have a gyroscope built-in to call for prompts from the program	Demonstration
1	COMP-01	—	The Arduino Uno R3 shall use a large breadboard, MPU9250, wires, HR-SR04, servo motor, potentiometer, and string	Inspection